ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- All-Pass filters (pp. 29-33) – two more slides have been added

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Digital Filter Fundamentals: All-Pass filters

All-Pass Filters

- An all-pass filter is one whose magnitude response is constant for all frequencies, but whose phase response is not identically zero.
- The simplest example of an all-pass filter is a pure delay system with system function $H(z) = z^{-k}$
- A more interesting all-pass filter is one that is described by

$$H(z) = \frac{a_L + a_{L-1}z^{-1} + \cdots + a_1 z^{-L+1} + a_0 z^{-L}}{1 + a_1 z^{-1} + \cdots + a_L z^{-L}}$$

where $a_0 = 1$ and all coefficients are real.

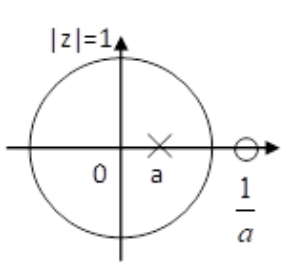
- If we define the polynomial $A(z)$ as

$$A(z) = \sum_{k=0}^{L} a_k z^{-k} \qquad a_0 = 1$$

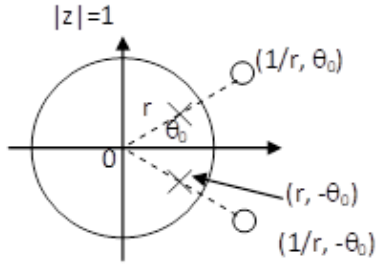
$$\therefore H(z) = z^{-L}\frac{A(z^{-1})}{A(z)}$$

- It can be shown that $|H(\theta)|^2 = H(z)\cdot H(z^{-1})|_{z=e^{j\theta}} = 1$ which is an all-pass filter.

- Furthermore, if $|z_0|$ is the modulus of a pole of $H(z)$, then $1/|z_0|$ is the modulus of a zero of $H(z)$ {i.e. the modulus of poles and zeros are reciprocals of one another}.
- The figure shown below illustrates typical pole-zero patterns for a single-pole, single-zero filter and a two-pole, two-zero filter.



All-pass



All pass filter

$$H(z) = \frac{1-\frac{1}{a}z^{-1}}{1-az^{-1}} \qquad |a| < 1 \text{ for stability}$$

- We can easily show that the magnitude response is constant.

$$|H(\theta)|^2 = H(\theta)\cdot H^*(\theta) = H(z)\cdot H(z^{-1})|_{z=e^{-j\theta}}$$

$$= \frac{1-\frac{1}{a}e^{-j\theta}}{1-ae^{-j\theta}}\cdot\frac{1-\frac{1}{a}e^{j\theta}}{1-ae^{j\theta}} = \frac{1-\frac{2}{a}\cos\theta+\frac{1}{a^2}}{1-2a\cos\theta+a^2} = 1/a^2$$

Digital Filter Fundamentals: All-Pass filters

Example 23:

- Show that the following transfer function $H_1(z)$ can be obtained using a parallel connection of a constant and an all-pass filter.

$$H_1(z) = \frac{10 + 6z^{-1}}{3 + z^{-1}}$$

Answer:

$$H_1(z) = \frac{9 + 3z^{-1} + 1 + 3z^{-1}}{3 + z^{-1}}$$

$$= 3 + \frac{1 + 3z^{-1}}{3 + z^{-1}}$$

$$H_1(z) = \underbrace{3}_{constant} + \underbrace{\frac{1 + 3z^{-1}}{3 + z^{-1}}}_{All-passfilter}$$

Example 24: All-pass filter

- Consider the filter described by

$H(z) = \frac{a+cz^2}{c+az^2}$ where a & c are constants.

Show that the magnitude response $|H(\theta)|$ is unity for all θ.

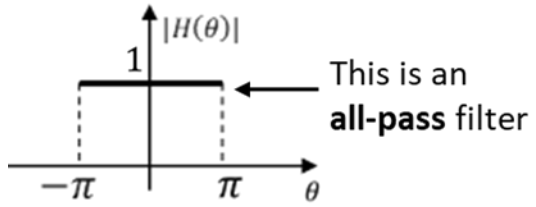
Answer:

$$H(z) = \frac{c+az^{-2}}{a+cz^{-2}}; \qquad H(\theta) = \frac{c+ae^{-j2\theta}}{a+ce^{-j2\theta}}$$

$$|H(\theta)|^2 = H(\theta) \cdot H^*(\theta)$$

$$= \frac{c + ae^{-j2\theta}}{a + ce^{-j2\theta}} \cdot \frac{c + ae^{j2\theta}}{a + ce^{j2\theta}}$$

$$= \frac{c^2 + a^2 + ac[e^{j2\theta} + e^{-j2\theta}]}{a^2 + c^2 + ac[e^{j2\theta} + e^{-j2\theta}]} = 1$$



Digital Filter Fundamentals: All-Pass filters

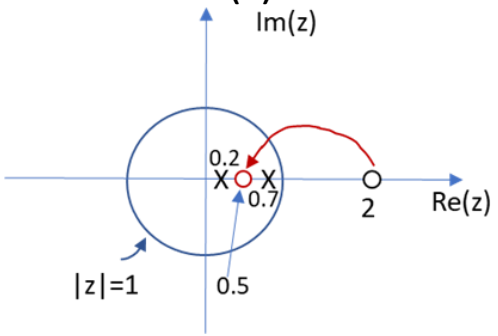
Example 25:

- A stable and causal non minimum phase filter may always be factored into a product of a minimum phase filter (with all poles and zeros inside the unit circle) with an all-pass filter.

$$H(z) = H_{min}(z).H_{allpass}(z)$$

- The procedure for performing this factorisation is as follows:
  - ❖ First, all of the zeros of H(z) that are outside the unit circle are reflected inside the unit circle to their conjugate reciprocal location. The resulting filter is minimum phase, $H_{min}(z)$.
  - ❖ Then, all-pass filter is selected so that it reflects the appropriate set of zeros of $H_{min}(z)$, back outside the unit circle.

- For the transfer function $H(z) = \frac{1-2z^{-1}}{(1-0.2z^{-1})(1-0.7z^{-1})}$, find the factorization for H(z) of the form $H(z) = H_{min}(z).H_{allpass}(z)$.



- The minimum phase factor is: $H_{min}(z) = \frac{1-(\frac{1}{2})z^{-1}}{(1-0.2z^{-1})(1-0.7z^{-1})}$
- Then , to reflect the zero at z=0.5 back outside the unit circle to z=2, we use the all-pass factor
- $H_{allpass}(z) = \underbrace{\frac{1-2z^{-1}}{1-(\frac{1}{2})z^{-1}}}_{All-passfilter}$

$$\therefore H(z) = \frac{1-(\frac{1}{2})z^{-1}}{(1-0.2z^{-1})(1-0.7z^{-1})} \cdot \frac{1-2z^{-1}}{1-(\frac{1}{2})z^{-1}}$$

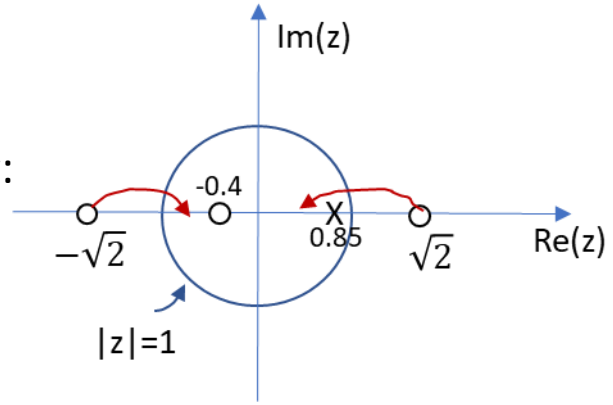
$$H(z) = H_{min}(z).H_{allpass}(z)$$

Digital Filter Fundamentals: All-Pass filters

Example 26:

- A causal LTI system has a system function: $H(z) = \frac{(1-2z^{-2})(1+0.4z^{-1})}{(1-0.85z^{-1})}$; Find a factorisation for H(z) of the form $H(z) = H_{min}(z).H_{allpass}(z)$, where $H_{min}(z)$ has minimum phase filter and $H_{allpass}(z)$ is a stable, all-pass, maximum phase filter .

The pole-zero map of H(z) is given by:



The system function H(z) has a nonminimum phase factor,$(1-2z^{-2})$, which may be written as the product of a minimum phase term and an all-pass factor as follows:

$$(1-2z^{-2}) = (1-\tfrac{1}{2}z^{-2}) \quad \frac{(1-2z^{-2})}{(1-\frac{1}{2}z^{-2})}$$

Therefore , H(z) may be written as the product of a minimum phase system with an all=pass system as follows:

$$H(z) = \underbrace{\frac{(1+0.4z^{-1})(1-\frac{1}{2}z^{-2})}{(1-0.85z^{-1})}}_{H_{min}(z)} \quad \underbrace{\frac{(1-2z^{-2})}{(1-\frac{1}{2}z^{-2})}}_{H_{allpass}(z)}$$